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A Model of the Regulatory Network Controlling Dendritic Cells Differentiation

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Abstract

Motivation: Dendritic cells have an important role during the immune response by acting as the main antigen-presenting cells. While there is experimental evidence about some key transcription factors and extracellular signals controlling their differentiation process, it is not yet clear the nature and dynamical properties of the regulatory network conformed by such molecules.
Results: We inferred the regulatory network that controls the differentiation process of murine dendritic cells. We then transformed the regulatory network into a discrete dynamical system, in the form of a Boolean model, to study its global properties. Our results are consistent with reported experimental expression patterns during the differentiation of these cells.
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1 Introduction

The vertebrate immune system generates both first-line response, as well as delayed protection against pathogens. Dendritic cells (DCs) constitute a central element linking the innate and the adaptive immune system. Originally, the description of DCs were based on their morphology, namely, as "large cells with cytoplasm arranged in a variety of cell shapes ranging from bipolar elongate cells stellate or dendritic ones" (Steinman and Cohn, 1973). Later functional studies resulted in a more modern classification with three main groups: classical DCs (cDCs), epidermal or Langerhans cells (LCs), and plasmacytoid dendritic cells (pDCs) (Schuler *et al.*, 1985; Colonna *et al.*, 2004). Gene expression data has yet to provide a distinctive molecular signature for each of the three DC types. In fact, there might exist a tissue-specific molecular signature of DCs even if they are classified as classical or plasmacytoid subtypes (Alcantara-Hernandez *et al.*, 2017). Broadly speaking, DCs express some hematopoietic markers such as CD45, MHC-II, FLT3, and CD11c, while at the same time lack markers corresponding to T, B, NKs, granulocytes and erythrocytes lineage markers (Merad *et al.*, 2013). Also, they express some transcription factors broadly found in other immune cells, such as IRF8, IRF4, Id2, Zbtb46, E2-2, Batf3, the STATs, RelB, Ikaros, Notch, RBP-J, and PU.1 (Perié and Naik, 2015). Additionally, DCs express a variety of pattern recognition receptor (PRRs), such as TLRs 1-11, and secrete some cytokines like IL-6, IL-10, and IL-12 (Manicassamy and Pulendran, 2009).

DCs have been the focus of much attention due to their potential therapeutic use, given that these cells are present at hematopoietic sites, environmental contact sites, filtering sites, and immune priming sites (Merad and Manz, 2009), as well as their ample repertoire of PRRs to recognize and uptake antigens, with the subsequent potential to stimulate specific T cell populations (Banchereau and Steinman, 1998; Shortman and Naik, 2007). These characteristics has prompted the efforts to generate *in vitro* particular DCs populations to generate specific adaptive responses (Harmon, 2011). For instance, some experimental approaches have been created against influenza and HIV infections (Boonnak *et al.*, 2013; Jacobson *et al.*, 2016), and recently there have been some promising results in the treatment of metastatic breast cancer (Zacharakis *et al.*, 2018).

DCs develop either from a common myeloid progenitor (CMP) or from a common lymphoid progenitor (CLP) (Manz *et al.*, 2001). FLT3L, which is produced by fibroblasts in the bone marrow, has been identified as the main external signal controlling the maturation process of DCs (Lisovsky *et al.*, 1996; Satpathy *et al.*, 2011; Murphy, 2016). Nonetheless, the precise sequence of molecular events during the differentiation of DCs still remain unknown.

Regulatory network modeling has been used successfully to study the global properties of several regulatory modules related to hematopoietic subprocesses (Mendoza and Méndez, 2015; Martinez-Sanchez *et al.*,

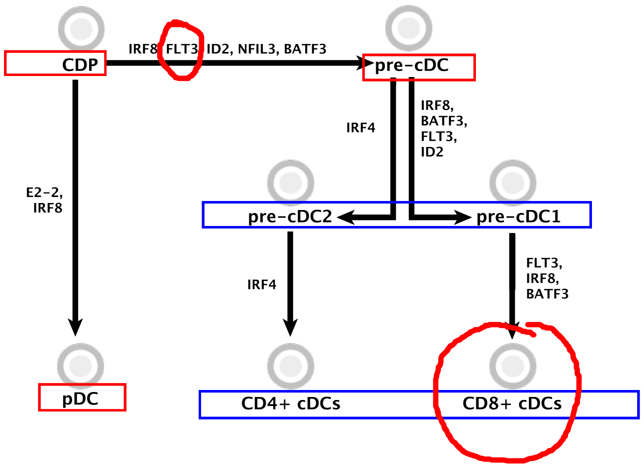

Fig. 1. Differentiation of dendritic cells. Arrows represent transitions from a progenitor to a more differentiated state. Names above arrows represent the main transcription factors reported during the process. Abbreviations are as follows: CDP, common dendritic progenitos. pre-cDC, pre- classical Dendritic Cell. pre-cDC1 and pre-cDC2, pre- classical Dendritic Cell type 1, and type 2, respectively. CD4+ cDCs and CD8+ cDCs, CD4+, and CD8+ classical dendritic cells respectively. pDC plasmacytoid dendritic cell.

2015; Méndez and Mendoza, 2016; Goode *et al.*, 2016; Hamey *et al.*, 2017; Ramirez and Mendoza, 2018; Liquitaya-Montiel and Mendoza, 2018). These studies have helped to understand how cells present specific molecular patterns under different physiological an experimental conditions as a direct result of the dynamical properties of an underlying regulatory network. We hereby present the regulatory network controlling the differentiation of DCs, inferred from a manual curation of experimental literature. The network is then converted into a Boolean model to study its global dynamical properties. The model recovers four attractors, which can be interpreted as the stable patterns of expression of the common dendritic progenitor (CDP), the pre-classical dendritic cell (pre-cDC), the pre-cDC type 1 (pre-cDC1), and the CD8+ classical dendritic cells (CD8+ cDCs). Furthermore, the model is able to describe the molecular patterns observed for single gain and loss-of-function mutations. Finally, we show that the model is able to describe the observed differentiation process from CDP to CD8+ cDCs.

64
= 62+2

2 Methods

2.1 CD8+ cDCs

The regulatory network controlling cDCs differentiation was inferred manually by reading a vast amount of experimental data available in the literature. Given the divesity of DCs types and sub-populations, and the varying amounts of molecular data available, it was decided to limit the study to the differentiation process of CD8+ cDCs splenic population in mice. The differentiation of CD8+ cDCs from a CMP is schematized in Fig. 1. These cells belong to DC1 DCs branch (Schlitzer *et al.*, 2015), they are tissue-resident, and their development is BATF3 and IRF8-dependent. Functionally, CD8+ cDCs recognize pathogens via TLR3, are capable of antigen cross-presentation, and stimulate CD8+ T cells (Shortman and Heath, 2010). It has been proposed that CD8+ cDCs murine have a human equivalent population defined as CD141+ (Villadangos and Shortman, 2010).

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2.2 Modeling framework

The regulatory network was converted into a dynamical system, specifically to a Boolean network, to study its global properties. A Boolean network is a set of nodes representing molecules or molecular complexes as well a set of regulatory directed interactions among them. In a Boolean network every node is described with a binary variable: 0 (OFF, repressed, or inactive) or 1 (ON, expressed, or active). The specific value of a node x_i is a function of the state of its regulators, so that $x_i(t+1) = F_i[x_1(t), x_2(t), ..., x_n(t)]$, where F_i is a Boolean function containing any combination of the logic operators AND, OR, and NOT. The vector containing the values of all nodes at a given time, i.e., $X(t) = x_1(t), ..., x_n(t)$, is the state of the network. Given the Boolean network has finite number of possible states, specifically 2^n, and that the system is deterministic, the trajectory created by successive iterations eventually reaches a previously visited network state. The trajectory consisting of a set of network states that repeat after n time steps is a period-n attractor. Period-1 attractors are also known as fixed points. Attractors represent phenotypes or cellular behaviors that can be directly compared to experimentally reported molecular patterns (Kauffman, 1969; Albert and Thakar, 2014).

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We first studied the dynamical behavior of the wild type network, and then we explored the behavior of all possible gain and loss-of-function single mutants. Mutants were simulated by fixing a specific node to a value of 0 to simulate a null-mutation, or to a value of 1 to represent a constitutive activation. The attractors of the wild-type and mutant models were obtained using the R package BoolNet 2.1.3 (Müssel *et al.*, 2010).

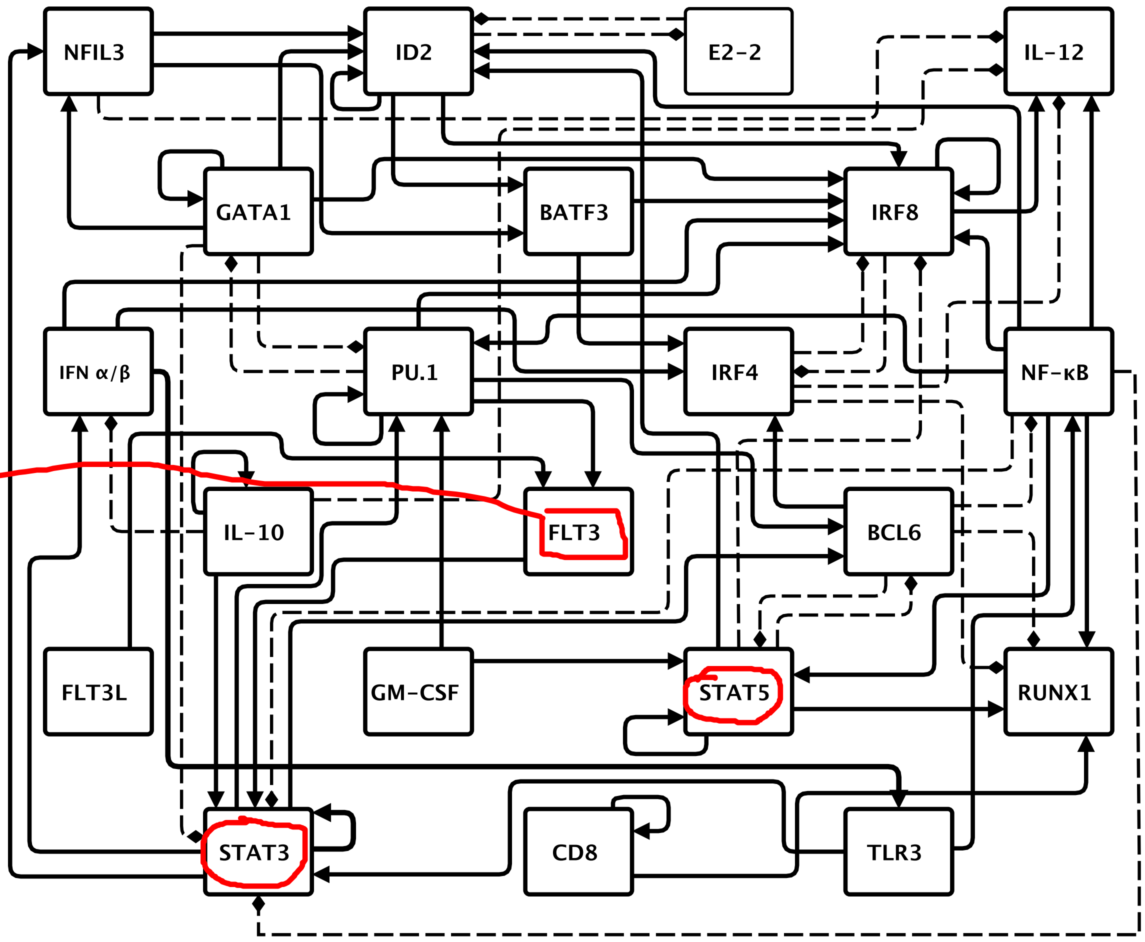
A cell fate map of the model was reconstructed by performing dynamical simulation starting the system at a given attractor, and then making a bit-flip on the state of one node and then letting the network converge to an attractor. This temporary change of state was performed systematically on all the nodes of the network, for all the attractors of the model. For clarity, only those perturbations that move the system from one attractor to another were reported.

3 Results

3.1 The molecular basis of the regulatory network

The regulatory network controlling differentiation of splenic CD8+ cDCs contains 21 nodes representing cytokines, receptors, transcription factors, and other molecular markers. There are a total of 64 regulatory interactions among the nodes, where 46 are activations and 18 inhibitions (Figure 2). Sixty-two interactions were inferred from experimental reports, detailed ahead, and it was necessary to incorporate two hypothetical regulatory interactions in order to recover the experimentally observed behavior.

The CD8+ cDCs differentiation process starts with the FLT3 receptor activation by FLT3L, which is secreted by fibroblasts and stromal cells in the bone marrow, as well as by spleen cells (Lisovsky *et al.*, 1996; Miloud *et al.*, 2012). FLT3 is expressed in CDP, pre-cDC preDC1 progenitors, and in the final differentiated state. Constant activation of FLT3L/FLT3 is essential for dendropoiesis (Hochweller *et al.*, 2009). Once activated, FLT3 signals downstream to regulate its target genes including STAT3 (Li and Watowich, 2012). STAT3 can auto-activate (Ichiba *et al.*, 1998), and also activate PU.1 (Laouar *et al.*, 2003), NFIL3 (Vallania *et al.*, 2009), IFN α/β (Moore *et al.*, 2001), and BCL6 (Walker *et al.*, 2013). PU.1 self-activates (Okuno *et al.*, 2005), and is a positive regulator of FLT3 (Carotta *et al.*, 2010), BCL6 (Wei *et al.*, 2009), and IRF8 (Schonheit *et al.*, 2013). ID2 might have a self-activating regulation (Florio *et al.*, 1998; Liu *et al.*, 2000), and also activates BATF3 and IRF8 (Jaiswal *et al.*, 2013). ID2 and E2-2 mutually inhibit each other, thus regulating cell fate; ID2 suppression by E2-2 drives the cell fate to pDCs, and E2-2 suppression by ID2 drives the cell fate to cDCs (Jackson *et al.*, 2011). BATF3 can be activated by NFIL3 (Kashiwada *et al.*, 2011), and in turn it activates



Start

21 nodes

Fig. 2. The regulatory network controlling the differentiation of splenic CD8+ cDCs in mice. Nodes (black squares) represent molecules. Sharp continuous arrows represent activations, while blunt discontinuous arrows represent inhibitions.

IRF8 and IRF4 (Grajales-Reyes *et al.*, 2015). IRF8 self-regulates positively (Kantakamalakul *et al.*, 1999; Bornstein *et al.*, 2014; Grajales-Reyes *et al.*, 2015) and has a mutual inhibition with IRF4 by which regulates the cell fate to CD8+ cDCs or CD4+ cDCs (Tamura *et al.*, 2005). Furthermore, both genes can be activated by IFN α/β (Guang-Nian *et al.*, 2015). In turn, IFN-α/β can be inhibited by IL-10 signalling (Ma *et al.*, 2015).

Experimentally, DCs can be generated using GM-CSF, which activates STAT5 and PU.1 at early differentiation states (Schiavoni *et al.*, 2002; Esashi and Liu, 2008). PU.1 has a mutual inhibition with GATA1, which in turn presents a self-activatory regulation (Zhang *et al.*, 1999; Nishimura *et al.*, 2000; Stopka *et al.*, 2005). GATA1 is able to activate IRF8 (Schonheit *et al.*, 2015), ID2 (Ji *et al.*, 2008), and NFIL3 (Choi *et al.*, 2015). Finally, NFIL3 is likely a positive regulator of ID2 (Xu *et al.*, 2015).

STAT3 and STAT5 have a central role during DCs differentiation (Onai and Manz, 2008). STAT3 is activated by TLR3, and is also part of the IL-10 signal transduction mechanism (Moore *et al.*, 2001; Williams *et al.*, 2004). Moreover, STAT3 is inhibited by GATA1 (Ezoe *et al.*, 2005) and by NF-κB. Regarding STAT5, it most likely has a self-activating regulation (Metser *et al.*, 2016), activates ID2 (Li *et al.*, 2012) and RUNX1 (Ogawa *et al.*, 2008), inhibits IRF8 (Esashi and Liu, 2008; Waight *et al.*, 2014), and presents a mutual inhibition with BCL6 (Walker *et al.*, 2013). In turn, BCL6 activates IRF4 (Ci *et al.*, 2009), and inhibits both RUNX1 (Hatzi *et al.*, 2015) and NF-κB (Perez-Rosado *et al.*, 2008).

Dendritic cell maturation involves pathogen recognition through the interaction between a pathogen-associated molecular pattern (PAMP) and a PRR. CD8+ cDCs express the toll-like receptor TLR3 induced by IFN α/β (Miettinen *et al.*, 2001). TLR3 activates the NF-κB response (Jiang *et al.*, 2004), which in turn positively regulates PU.1 (Bonadies *et al.*, 2009), STAT5 (Katerndah *et al.*, 2017), ID2 (Wang *et al.*, 2012), IRF8 (Simon *et al.*, 2015), RUNX1 (Zhou *et al.*, 2015; Tang *et al.*, 2018), and IL-12 (T.L. *et al.*, 1995; Grohmann *et al.*, 1998; Hoentjen *et al.*, 2005), while inhibiting STAT3 (Ahmad *et al.*, 2015). RUNX1 is inhibited by IRF4 (Cao *et al.*, 2010). In turn, IL-12 is negatively regulated by IRF4 and NFIL3, but positively regulated by IRF8 (Akbari *et al.*, 2014; Kobayashi *et al.*, 2014). Finally, IL-12 can also be negatively regulated by IL-10 in a autocrine or niche-depending-way (Rennick *et al.*, 1997; Corinti *et al.*, 2001), and is self-regulated positively (Knolle *et al.*, 1998).

Despite the amount of published experimental data, we had to make some hypotheses regarding the existence of some regulatory interactions. Specifically, since there was no available information regarding the molecular basis of the regulation of CD8 expression, we included a self-activation, as well as its positive regulation over RUNX1, as originally proposed by a previous regulatory network model (Mendoza and Méndez, 2015).

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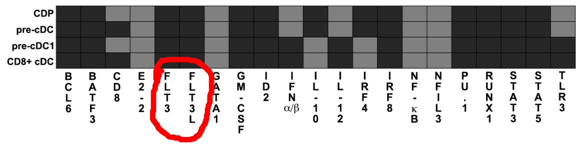

Fig. 3. Attractors of the Boolean model. Wild-type model attractors. Active nodes are marked in black, and inactive nodes in gray. The corresponding cell types are indicated in the leftmost column.

3.2 Dynamics of the regulatory network

We constructed a Boolean model by assigning logical functions describing the activation of individual nodes as a function of its regulators (Table 1). Notice that two nodes, FLT3L and GM-CSF, act as input to the network. Either of these molecules are necessary for the differentiation of Dendritic Cells. Therefore, during the modeling process these two nodes are always active, unless stated otherwise. An exhaustive analysis of the resulting discrete dynamical system showed that the network has only four attractors, each of these being fixed-points that correspond to the general molecular activation patterns observed experimentally for the cell types CDP, precDC, pre-cDC1, and CD8+ cDCs (Figure 3).

There are small discrepancies between the model and the experimental data. Specifically, it has been reported that BCL6 expression is detected starting at the pre-cDC progenitor (Zhang *et al.*, 2014). In the model, however, BCL6 is detected also at the CDP stage. This is an indication that the model can be improved by looking maybe for a negative regulator. Then, it has been suggested that NFIL3 expression is restricted to CDP or pre-DC states (Kashiwada *et al.*, 2011; Merad *et al.*, 2013; Murphy *et al.*, 2016), while in the attractors this node is always in its off state. The model is thus probably missing a positive regulation over NFIL3.

Despite the relative minor discrepancies mentioned above, the model describes qualitatively the stationary expression patterns of key cell types during the differentiation of CD8+ cDCs. Furthermore, the model is able to qualitatively reproduce the differentiation process itself. CDPs are induced to differentiate by the constant presence of the extracellular signal FLT3L in the bone marrow or spleen, or by constant GM-CSF in vitro (Karsunky *et al.*, 2003; Xu *et al.*, 2007; Conti and Gessani, 2008; Helft *et al.*, 2015). An initial developmental decision is the commitment to either the pDC or cDC cell fates. This lineage separation is regulated by the mutual inhibition between ID2 and E2-2 (Belz and Nutt, 2012). The fate map (Figure 4) shows that in the model, under the constant input of either FLT3L or GM-CSF the activation of either BATF3 or ID2 moves the system from the CDP attractor to the pre-cDC attractor. It is worth mentioning that there might be a synergistic effect between ID2 and BATF3 during the differentiation process (Jaiswal *et al.*, 2013). Moreover, notice than in the fate map the presence of BATF3 is able to move the system from CDP trough pre-cDC and pre-CD1 and finally to cDC CD8+. This characteristic supports the idea that these cells might be characterized as BATF3-depending cDCs (Murphy *et al.*, 2016).

IL-10 repression has been recognized as a maturation indicator during DC differentiation (Morel *et al.*, 1997). This event is also qualitatively recovered in the model, as the fate map shows that the inactivation of IL-10 moves the system from pre-cDC to pre-cDC1. Experimentally reports, however, suggest a gradual loss of sensibility of DCs to IL-10 (MacDonald *et al.*, 1999), thus preventing spontaneous activation of T cells (Corinti *et al.*, 2001).

Finally, results shown on the fate map suggest that the inactivation of BCL6 blocking, or the activation of E2-2 might move a pre-cDC to a CDP-like attractor (Figure 4) which exhibit the potential to differentiate to pDC or cDC branches given the expression of E2-2, ID2 and STAT3 but with the absence of NFIL3. This transition can be interpretes as a

Table 1. The logic rules determining the activity of all modes in the network.

NODE	REGULATORY INTERACTIONS
BCL6	(STAT3 OR PU.1) AND NOT STAT5
BATF3	ID2 OR NFIL3
CD8	CD8
E2-2	NOT ID2
FLT3	FLT3L AND PU.1
FLT3L	is an input
GATA1	GATA1 AND NOT PU.1
GM-CSF	is an input
ID2	(NFIL3 OR STAT5 OR NF-κB) AND NOT E2-2
IFN α/β	STAT3 AND NOT IL-10
IL-10	IL-10
IL-12	(NF-κB AND IRF8) AND NOT IL-10 AND NOT NFIL3
IRF4	BATF3 AND BCL6 OR (IFN α/β AND NOT IRF8)
IRF8	(IRF8 OR IFN α/β OR BATF3 OR NF-κB OR PU.1 OR ID2 OR RUNX1 OR GATA1) AND NOT IRF4 AND NOT STAT5
NF-κB	TLR3 AND NOT BLC6
NFIL3	STAT3 AND GATA1
PU.1	(PU.1 OR GM-CSF OR STAT3 OR NF-κB) AND NOT GATA1
RUNX1	(CD8 OR STAT5 OR NF-κB) AND NOT IRF4 AND NOT BCL6
STAT3	(STAT3 AND TLR3) OR (FLT3 OR IL-10) AND NOT NF-κB AND NOT GATA1
STAT5	(STAT5 AND GM-CSF OR NF-κB) AND NOT BCL6
TLR3	IFN α/β

de-differentiation, from a more differentiated state to a progenitor or less differentiated state. This transition has not been reported experimentally, thus it is prediction of the present model. ?

3.3 Simulation of mutants and perturbations

We analyzed all possible single loss- and gain-of-function mutants of the model, however only those with biological relevance are reported and discussed in the main text (see Table 2). The attractors of all the simulated mutants are reported in the Supplementary Information. Importantly, models BCL6(+), BATF3(+), E2-2(-), FLT3(+), FLT3L(-), GATA1(-), ID2(+) IRF8(+), NF-κB(-), NFIL3(-), PU.1(+), RUNX1(+), STAT3(+), and SATAT5(+) show no differences with respect to wild-type model in terms of the attained attractors.

FLT3L and GM-CSF act as partially redundant inputs to the network. Thus, eliminating only one signal does not affect the behavior of the model, but the double negative mutant FLT3L(-)/GM-CSF(-) does (Supplementary Table 43), where all the attractors are lost.

BCL6(-) mutant model shows a complete repression of IRF4 (Supplementary Table 1), but otherwise the resulting attractor are very similar to those obtained in the wild-type model. Experimentally, it is that IRF4 has an essential role during late CD4+ cDCs differentiation. Therefore, the lack of a major effect in our model is consistent with reports on the decreasing but not complete ablation of CD8+ cDCs number in BCL6 knockout models (Ohtsuka *et al.*, 2011). A somewhat similar effect is observed in the BATF3(-) mutant model (Supplementary Table 3), showing a repression of both IRF4 and IL-12. This might correlate with the experimental observation that BATF3 suppression results in an inefficient stimulation of CD8+ cDCs, rather than a reduction in their number (Montagna *et al.*, 2015).

PU.1 is involved the differentiation process of some hematopoietic lineages, particularly at the level of progenitors (Fisher and Scott, 1998; Iwasaki *et al.*, 2005). In the model, the PU.1(-) mutant results in the loss of all attractors (Supplementary Table 33), which is consistent with the report that this mutant shows impaired DC formation *in vitro* (Carotta *et al.*, 2010).

The IRF8(-) mutant model (Supplementary Table 27) has an extra attractor somewhat similar to that expected for CD4+ cDCs, characterized by the FLT3L/FLT3 dependence, the presence of ID2 and IRF4, as well as the absence of IL-12 production, and the expression of TLR3. However, given the some of the key molecular markers of these cells (such as NOTCH

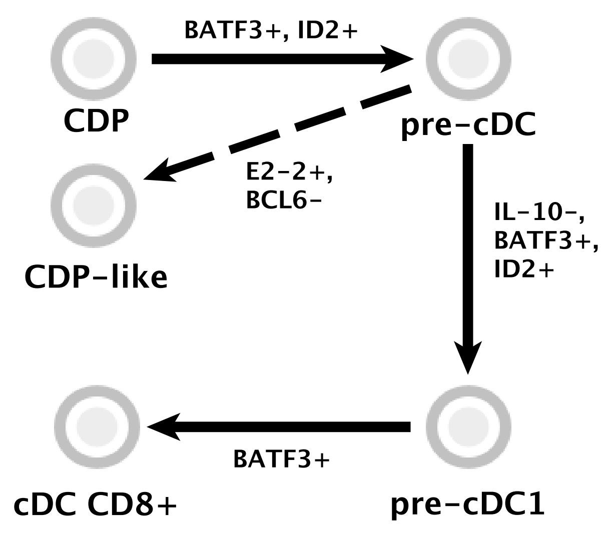

Fig. 4. Fate map. Arrows represent transition between a progenitor cellular sate into a more differentiated state, and names above represent possible single node effects leading these transitions. Dashed arrow represent theoretical possible transition (see discussion section).

Stat3 loss

any double mutation very human?

Table 2. Some relevant mutants

Mutant	Effect on the model	Experimental counterpart
IRF8 loss	cDC CD4-like attractor	Murphy et al., 2016
FLT3L loss	No effect	Karsunky et al., 2003
NFIL3 gain	No IL-12	Kashiwada et al. 2011
PU.1 loss	Unspecific attractors	Carotta et al., 2010
BCL6 loss	IRF4 repression	Ohtsuka et al., 2011
BATF3 loss	IRF4 repression, no IL-12	Montagna et al., 2015
ID2 loss	pDC-like attractor	Ghosh et al., 2010
STAT5 loss	pDC-like atractor	Li et al., 2012

? only 1 reference ?

signalling, some IRF4 regulators, IL-23 production, etc.) are not included in the present version of the the model, it is not possible to unequivocally describe a CD4+ attractor. ?

The ID2(-) and STAT5(-) mutant models (Supplementary Tables 17 and 39) reveal a pDC-like attractor characterized by the absence of cDC-related regulators IRF4, BATF3, NFIL3, no IL-12 production and no TLR3 expression, but with the FLT3L/FLT3 dependence, and STAT3, IRF8 and E2-2 presence. It is known that ID2 and STAT5 suppression leads the E2-2 and STAT3 action, both related to pDC differentiation program (Cisse *et al.*, 2008; Li *et al.*, 2012).

4 Discussion

DCs are a key component of the cellular immune response, but the regulatory mechanisms controlling their differentiation remains unclear. We used publicly available information to reconstruct a regulatory network involved in the differentiation of splenic CD8+ DC from the common dendritic progenitor. The model of the network as a Boolean model recovers the expression patterns observed in CDPs, pre-cDCs, pre-cDC1s, and CD8+ cDCs. While there have been previous efforts to uncover the molecular networks associated to dendritic cells (see, for example Pandey *et al.* (2013); Lin *et al.* (2015)), to the best of our knowledge this is the first network of its kind analyzed as a dynamical system to describe the differentiation process of CD8+ cDCs.

The definition of DC lineages has been cause of controversy for many years, given that they may have either a myeloid or a lymphoid origin. Accordingly, there have been some attempts to redefine DCs lineages based on specific molecular characteristics (Perié and Naik, 2015; Murphy *et al.*, 2016). The network presented here contains nodes that are shared with other myeloid and lymphoid lineages (Mendoza and Méndez, 2015; Méndez and Mendoza, 2016; Ramirez and Mendoza, 2018; Liquitaya-Montiel and Mendoza, 2018), and hence it allows for its integration into a more complex network that may help in the understanding of the different possible differentiation routes to generate dendritic cells during hematopoiesis.

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